

Vector Spaces, Bases, Rank and Projections

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5.1 What a subspace represents

A vector space over the real numbers is a set equipped with addition and scalar multiplication satisfying the usual vector-space axioms. In this chapter we work mainly in finite-dimensional spaces of real coordinate vectors. A subspace is a nonempty subset closed under both operations. It contains the zero vector: a shifted plane that does not pass through the origin is not a subspace.

The span of vectors $v_1, \dots, v_k$ is the set of all their real linear combinations. If the vectors are stored as the columns of a matrix B , their span is the column space of B .

$$\text{span}\{v_1, \dots, v_k\} = \{c_1 v_1 + \dots + c_k v_k : c_1, \dots, c_k \in \mathbb{R}\}$$

Worked example: membership in a plane

Take the columns $(1, 0, 0)^T$ and $(0, 1, 0)^T$. Every combination has third coordinate zero. Thus $(2, 3, 0)^T$ lies in the span, with coefficients 2 and 3, whereas $(2, 3, 1)^T$ does not. The notebook verifies both membership statements numerically.

Learning outcomes

By the end of the chapter, you should be able to distinguish span, independence, basis and dimension; compute coordinates in a basis; explain column, row and null spaces; construct and test an orthogonal projector; and interpret least squares and model residuals geometrically.

5.2 Independence, bases and coordinates

Vectors are linearly independent when the only linear combination equal to zero has every coefficient equal to zero. A basis is an independent spanning set. The dimension of a finite-dimensional vector space is the number of vectors in any of its bases. Adding a redundant vector to a spanning set does not increase the dimension.

For example, the columns of the 2×2 identity matrix are independent. The columns $(1, 2)^T$ and $(2, 4)^T$ are dependent because the second is twice the first. Their matrix has rank 1, not 2.

Worked example: one vector, two descriptions

Let B have columns $b_1 = (1, 1)^T$ and $b_2 = (1, -1)^T$. For $v = (4, 2)^T$, its coordinate vector c in basis B is determined by $Bc = v$. The notebook denotes this coordinate vector by $[v]$ with the basis label B as a subscript.

$$\begin{aligned} c_1 + c_2 &= 4, & c_1 - c_2 &= 2 & \Rightarrow & c = (3, 1)^T \\ v &= 3b_1 + b_2 = (4, 2)^T \end{aligned}$$

The vector itself has not changed. Only its description has changed from standard coordinates $(4, 2)$ to basis coordinates $(3, 1)$. The notebook checks reconstruction, not just the printed coefficient values.

Changing between two bases

For two bases of $\mathbb{R}^n$, arrange their vectors in nonsingular square matrices B_1 (old basis) and B_2 (new basis). Equality of the reconstructed vector gives $B_1c_1 = B_2c_2$. Solve the system below for the change-of-coordinate matrix T rather than explicitly computing an inverse in numerical code.

$$B_2T = B_1, \quad c_2 = Tc_1$$

From the standard basis to B in the example, T has rows $(1/2, 1/2)$ and $(1/2, -1/2)$. It sends $(4, 2)^T$ to $(3, 1)^T$. A rectangular full-column-rank matrix can represent a basis of a smaller subspace, but the notebook's change-of-basis routine deliberately requires square bases of the ambient space.

Coordinates are unique only when the supplied spanning columns are independent and the vector lies in their span. A least-squares coefficient vector is not automatically an exact, unique coordinate representation.

5.3 Rank and the fundamental subspaces

Let A be an $m \times n$ real matrix, viewed as a map from $\mathbb{R}^n$ to $\mathbb{R}^m$. The column space lies in $\mathbb{R}^m$ and contains every attainable output Ax . The row space is the span of the rows in $\mathbb{R}^n$. The null space also lies in $\mathbb{R}^n$ and consists of all inputs x satisfying $Ax = 0$.

$$\text{rank}(A) = \dim \text{Col}(A) = \dim \text{Row}(A)$$

$$\text{rank}(A) + \dim \text{Null}(A) = n$$

Rank-nullity divides the input dimension n into independent directions detectable by the map and directions annihilated by it. The row space and null space are orthogonal complements in the input space. The column space and null space must not be treated as subspaces of the same ambient space when m and n differ.

Worked example: a 2×3 matrix

$$A = [(1, 2, 3); (2, 4, 6)]$$

In this compact matrix notation, semicolons separate rows. The second row is twice the first, so $\text{rank}(A) = 1$. Its column space is spanned by $(1, 2)^T$ and its row space by $(1, 2, 3)$. The equation $Ax = 0$ reduces to $x_1 + 2x_2 + 3x_3 = 0$.

$$x = s(-2, 1, 0)^T + t(-3, 0, 1)^T, \quad s, t \in \mathbb{R}$$

The null space therefore has dimension 2, and $1 + 2 = 3$ confirms rank-nullity. The notebook obtains orthonormal bases using singular value decomposition (SVD). Those bases may have different signs or entries from the simple basis above, yet span exactly the same spaces.

Numerical rank is a modeling decision too

In floating-point computation, a singular value is treated as zero relative to a chosen tolerance. The candidate library uses a scale-dependent threshold based on the largest singular value, machine precision and matrix dimensions. Nearly dependent columns can consequently be classified differently under different tolerances. Record tolerance choices when results are sensitive to them.

Do not compare basis arrays entry by entry to prove equality of subspaces. Check reconstruction, orthogonality or the corresponding projectors. For a computed null-space basis N , a useful numerical check is that AN is close to zero.

5.4 Orthogonal projections

An orthogonal projection separates a vector x into a component p in a chosen subspace S and a residual r perpendicular to every vector in S . The projected point is the closest point in S in Euclidean distance.

$$x = p + r, \quad p \in S, \quad r \perp S$$

If Q has orthonormal columns spanning S , then $Q^T x$ gives coordinates along those directions and $Q(Q^T x)$ reconstructs the projected component. The matrix P is the orthogonal projector.

$$P = QQ^T, \quad p = Px, \quad r = x - Px$$

$$P^T = P, \quad P^2 = P, \quad Q^T r = 0$$

Symmetry and idempotence together characterize real orthogonal projectors. Idempotence alone is not enough: an oblique projection can satisfy $P^2 = P$ without being symmetric or producing a perpendicular residual.

Worked example: projection onto the coordinate plane

For the plane spanned by $(1, 0, 0)^T$ and $(0, 1, 0)^T$, $P = \text{diag}(1, 1, 0)$. Given $x = (2, 3, 4)^T$, the projected component is $(2, 3, 0)^T$ and the residual is $(0, 0, 4)^T$. The residual is perpendicular to both spanning directions. Their squared lengths satisfy $29 = 13 + 16$.

The notebook plot shows only the first and third coordinates. Its horizontal arrow therefore represents $(2, 0)$ in that two-coordinate view, not the full three-dimensional projected vector. The omitted second coordinate remains 3.

Nonorthonormal and redundant spanning columns

For a full-column-rank matrix B , the same projector is $B(B^T B)^{-1} B^T$. This formula explains the geometry but explicitly forming the inverse is usually undesirable. If B has dependent columns, $B^T B$ is singular and that formula cannot be used as written.

The candidate implementation takes the retained left singular vectors of B as Q . This also handles zero or dependent leading columns and the zero subspace. Truncating an unpivoted QR factorization merely by the matrix rank is not reliable for such input. The added regression tests cover that failure case.

5.5 Least squares, statistics and AI

When $Ax = b$ cannot be solved exactly, least squares chooses coefficients β to minimize the squared Euclidean residual. The fitted vector $A\beta$ is the orthogonal projection of b onto the column space of A .

$$\text{minimize } \|b - A\beta\|^2, \quad r = b - A\beta, \quad A^T r = 0$$

The fitted vector is unique because the orthogonal projection onto a fixed subspace is unique. The coefficients are unique only when A has independent columns. Rank-deficient designs may admit several coefficient vectors with the same fitted values. NumPy's least-squares solver selects a minimum-norm solution under its numerical rank convention.

Worked example: fitting an intercept and a slope

Use observations at $t = 0, 1, 2$ with responses $b = (1, 2, 2)^T$. Let A have rows $(1, 0), (1, 1), (1, 2)$, so β contains the intercept and slope. The normal equations provide a hand calculation:

$$3\beta_0 + 3\beta_1 = 5, \quad 3\beta_0 + 5\beta_1 = 6$$

$$\beta = (7/6, 1/2)^T$$

$$A\beta = (7/6, 5/3, 13/6)^T, \quad r = (-1/6, 1/3, -1/6)^T$$

The residual sums to zero and its inner product with $(0, 1, 2)^T$ is also zero. Thus it is orthogonal to both design columns. Its squared norm is $1/6$. The notebook solves with `numpy.linalg.lstsq`; using the normal equations here is an explanatory hand calculation, not a recommendation to use them for poorly conditioned numerical problems.

Where this geometry appears

Regression uses design-column spaces. PCA projects centered data onto selected orthonormal principal directions. Low-rank representations restrict data to a lower-dimensional model, retaining some variation and discarding the remainder. Interpretation requires knowing the chosen metric, preprocessing and subspace.

In attention architectures, the word “projection” often means a learned linear transformation of embeddings. Such a map need not be square, symmetric or idempotent. It should not be described as an orthogonal projection unless those additional mathematical conditions actually hold.

5.6 A policy subspace: diagnostic, not verdict

Suppose three standardized coordinates summarize Agriculture, Health and Energy priorities. Choose illustrative archetypes $b_1 = (1, 0.2, 0.6)^T$ and $b_2 = (0.2, 1, 0.6)^T$. They are independent and define a two-dimensional subspace. A proposed profile is $x = (0.9, 0.7, 0.1)^T$. All values are illustrative, not empirical estimates.

$$p = (2/3, 7/15, 17/30)^T$$

$$r = x - p = (7/30, 7/30, -7/15)^T$$

The residual is perpendicular to both archetypes. A least-squares representation of p uses coefficients $43/72$ and $25/72$ on b_1 and b_2 respectively. The notebook displays the proposed, aligned and residual components and checks orthogonality.

The negative Energy residual says that the proposal gives less weight to that coordinate than its fitted subspace component does. It does not establish that the proposal is wrong. Conclusions depend on the archetypes, scaling and metric. Linear subspaces also allow negative coefficients; if policy weights must be nonnegative or sum to one, use an explicitly constrained model rather than assuming an unconstrained projection respects those requirements.

Practice with worked answers

A. Is the set of vectors $(a, b, 1)^T$ a subspace of $\mathbb{R}^3$? No: it excludes zero and is not closed under scalar multiplication. The set $(a, b, 0)^T$ is a subspace, of dimension 2.

B. Find coordinates of $(6, 2)^T$ in the basis $(1, 1)^T, (1, -1)^T$. Solve $c_1 + c_2 = 6$ and $c_1 - c_2 = 2$, giving $(4, 2)^T$. Reconstruct to check.

C. Project $(2, 3)^T$ onto the line spanned by $(1, 1)^T$. The coefficient is $5/2$, so $p = (5/2, 5/2)^T$ and $r = (-1/2, 1/2)^T$. Their inner product is zero. The projector has every entry equal to $1/2$.

D. Explain the runtime regression case $B = [(0, 0); (0, 1)]$. Its column space is the second coordinate axis, so the projector must be $\text{diag}(0, 1)$, regardless of column order. This is why the candidate uses SVD rather than truncating unpivoted QR.

Further study and control boundary

Further reading: Gilbert Strang, Introduction to Linear Algebra; David C. Lay, Steven R. Lay and Judi J. McDonald, Linear Algebra and Its Applications; Lloyd N. Trefethen and David Bau III, Numerical Linear Algebra. Companion notebook: v0.2.2. Owner approval covers the chapter and notebook; independent sign-off is not recorded. Runtime 1.1.1rc1 remains a prerelease. Publication requires separate authorization.